

Adaptive Friction Velocity Verlet Optimization for Variational Quantum Eigensolvers

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Abstract - The Variational Quantum Eigensolver (VQE) represents a leading hybrid quantum-classical algorithm tailored for contemporary quantum hardware. Nonetheless, the optimization of parameterized quantum circuits presents a significant challenge due to intricate, non-convex energy landscapes. Recently, optimizers inspired by classical physics, such as the Velocity Verlet method, have been introduced to effectively traverse these landscapes by utilizing simulated momentum. This paper introduces an "Adaptive Velocity Verlet" (AVV) approach that incorporates energy-aware dynamic time-stepping along with exponential momentum decay. By persistently tracking the trajectory of the cost function, the AVV algorithm adjusts the integration step size dynamically and imposes an emergency momentum penalty when energy overshoots are detected. We evaluate the performance of the proposed AVV against the conventional Velocity Verlet, L-BFGS-B, SLSQP, and COBYLA methods, employing Qulacs and PySCF statevector simulators for Hydrogen H_2 and Lithium hydride (LiH) molecules. The results reveal a distinct trade-off: while AVV decreases the total number of quantum circuit evaluations (18,281 compared to 19,241 for LiH), thereby enhancing computational efficiency, it compromises ground-state accuracy by becoming ensnared in local minima during periods of aggressive momentum decay.

I. INTRODUCTION

A. The Promise of Near-Term Quantum Computing

The present state of quantum computing is characterized by the Noisy Intermediate-Scale Quantum (NISQ) era. NISQ hardware is limited in the number of qubits it can utilize and does not possess the extensive logical depth necessary for comprehensive quantum error correction. As a result, hybrid quantum-classical algorithms have surfaced as the leading framework for realizing practical quantum applications prior to the development of fault-tolerant systems. The Variational Quantum Eigensolver (VQE) is recognized as the principal algorithm in this field, primarily aimed at estimating the ground-state energy of molecular Hamiltonians relevant to quantum chemistry and materials science [2], [6]. The VQE functions based on the variational principle. It assigns the task of preparing a parameterized trial state (the ansatz) and measuring the expectation value of the Hamiltonian to the quantum processing unit (QPU). Subsequently, the classical computer is responsible for iteratively adjusting the ansatz parameters to reduce the cost function (the energy).

B. The Classical Optimization Bottleneck

Although the VQE is theoretically sound, its scalability is significantly limited by the classical optimization stage. As the number of qubits rises to represent larger molecular systems, the parameter space of the quantum circuits becomes increasingly intricate. These spaces frequently display rugged, non-convex energy landscapes marked by a multitude of shallow local minima. Additionally, researchers face the challenge of "barren plateaus," which are extensive, flat areas in the optimization landscape where the gradients of cost function diminish exponentially as number of qubits rise, making gradient-based search technique ineffective [5].

C. Contribution of this Paper

In order to traverse these intricate quantum terrains, it is essential to have optimizers that can effectively balance extensive exploration with focused exploitation. This study presents a new improvement to optimization algorithms inspired by physics. In particular, we propose an "Adaptive Velocity Verlet" (AVV) approach that incorporates real-time, energy-sensitive dynamic time-stepping and exponential momentum decay to avoid algorithmic trapping and overshooting during VQE optimization processes.

II. LITERATURE REVIEW

The challenge of optimizing parameterized quantum circuits has generated a vast body of literature, with researchers actively exploring both gradient-free and gradient-based classical algorithms, as well as novel physics-inspired approaches.

A. Traditional Heuristic and Gradient-Based Optimizers

Initial versions of the Variational Quantum Eigensolver (VQE) predominantly utilized gradient-free, heuristic optimization techniques. Algorithms such as Nelder-Mead and COBYLA (Constrained Optimization BY Linear Approximations) have demonstrated significant effectiveness for very small systems, such as single-qubit or two-qubit setups. Nonetheless, as highlighted in recent studies on scalability, these gradient-free approaches encounter difficulties in converging within high-dimensional parameter spaces, often necessitating an impractical number of evaluations of quantum circuits.

In contrast, gradient-based optimization techniques like L-BFGS-B (Limited-memory Broyden Fletcher Goldfarb Shanno) and SLSQP (Sequential Least Squares Programming) provide mathematically sound solutions with

quicker convergence rates. However, despite their efficiency, these methods are inherently greedy. In the realm of VQE, where the optimization landscape is fraught with sub-optimal energy wells, gradient-based techniques are particularly prone to becoming ensnared in local minima, thus failing to achieve the global ground state essential for chemical precision. Additionally, their vulnerability to barren plateaus is a well-documented drawback.

B. Stochastic and Momentum-Based Approaches

To overcome the shortcomings of traditional gradient descent, the quantum computing sector has incorporated optimizers from the realm of deep learning. Techniques like Adam (Adaptive Moment Estimation) and SPSA (Simultaneous Perturbation Stochastic Approximation) are commonly implemented. SPSA is particularly advantageous in environments with noise, as it estimates the gradient by using only two energy evaluations per iteration, independent of the number of parameters involved. Although momentum-based techniques such as Adam are more effective at avoiding local minima compared to standard gradient descent, they still necessitate careful hyperparameter tuning (learning rate scheduling), which must be tailored to each specific molecule and ansatz architecture.

C. Physics-Inspired Optimization Methodologies

Acknowledging the constraints of conventional classical optimizers within quantum landscapes, contemporary research has shifted towards algorithms inspired by physics. A particularly noteworthy and promising strategy involves the use of classical molecular dynamics simulations for quantum parameter updates, specifically through the implementation of Velocity Verlet integration [1]. In this context, the quantum parameters are conceptualized as "particles" traversing a physical state-space, where the gradient of the cost function serves as a physical force that propels the particles down the energy gradient. The traditional Velocity Verlet method is adept at accumulating simulated kinetic momentum, enabling the optimizer to navigate through shallow local minima that might otherwise ensnare an algorithm such as L-BFGS-B.

D. Identifying the Literature Gap

While the Velocity Verlet approach demonstrates superior momentum-gathering capabilities, current literature highlights a significant flaw: the reliance on static, hardcoded hyperparameters. The base implementations utilize a fixed integration time-step and a constant damping friction. This rigidity often causes the simulated particles to accumulate excess kinetic energy without a mechanism to bleed it off. Consequently, the optimizer experiences severe oscillations around the global minimum or violent ejections from optimal energy wells. There remains a critical gap in the literature for an adaptive framework capable of actively modulating these physical parameters in real-time based on landscape topology, which this study directly addresses.

III. METHODOLOGY

A. Quantum Hamiltonian Formulation

The experiments were constructed using the PySCF quantum chemistry package [4] paired with the Qulacs high-performance statevector simulator [3]. Two molecular systems were benchmarked: Hydrogen (H_2) at a bond distance of 0.977 Å, requiring 4 qubits, and Lithium hydride (LiH) at a bond distance of 1.596 Å, requiring 12 qubits. The molecular Hamiltonians were generated using the STO-3G minimal basis set, and the fermionic operators were mapped to qubit operators via the Jordan-Wigner transformation.

B. Hardware-Efficient Ansatz

A Hardware-Efficient Ansatz (HEA) was utilized to construct the parameterized quantum state. The circuit depth was established at $D=4$ for both molecules. This ansatz comprises layers of parameterized single-qubit rotations (RY and RZ gates) interspersed with entangling layers that include controlled-Z (CZ) gates applied to neighboring qubits. The expectation value of the Hamiltonian corresponding to this state acts as the cost function that needs to be minimized.

C. Standard Velocity Verlet Optimizer

The foundational methodology employs the traditional Velocity Verlet integration technique. Let θ denote the parameters of the quantum circuit, v signify the artificial velocity, m indicate a hypothetical mass, and γ represent the damping coefficient. The rule for updating the discrete time-step is articulated as follows:

- I. Determine the initial force through analytical gradients.
- II. Revise the provisional velocity.
- III. Adjust the position (parameters) utilizing the provisional velocity and the time-step.
- IV. Compute the new force at the updated position.
- V. Conclude the velocity by applying the damping coefficient to the new force and the provisional velocity.

D. Proposed Methodology: Adaptive Velocity Verlet (AVV)

In order to tackle the issues of oscillation and overshoot associated with the conventional method, this paper presents an energy-aware dynamic time-stepping and momentum decay protocol known as Adaptive Velocity Verlet (AVV). The AVV algorithm incorporates a continuous monitoring condition of the landscape immediately after the provisional parameter update.

Algorithm Rule 1 (Overshoot Detection and Braking):

When the provisional energy exceeds the current energy, the algorithm identifies an unwanted rise in the energy landscape (an overshoot). Consequently, the update is rejected, and an emergency exponential penalty is enforced, resulting in a 90% decay of the velocity and a halving of the integration step size. To ensure computational efficiency during a braking event, the analytical gradient from the preceding step is stored and reused, thereby avoiding unnecessary and costly evaluations of the quantum circuit.

Algorithm Rule 2 (Smooth Terrain Acceleration):

If the provisional energy is less than or equal to the current energy, the update is permitted. To enhance convergence in flat regions, the integration step size is cautiously increased by 5%, with a maximum limit imposed.

IV. RESULTS AND DISCUSSION

The proposed AVV methodology was benchmarked against the standard Velocity Verlet, L-BFGS-B, SLSQP, and COBYLA. The primary metrics of evaluation were the final absolute error relative to the Full Configuration Interaction (FCI) exact energy, and the total number of quantum energy evaluations (a direct measure of computational cost). The full performance comparison is summarized in Table I.

Optimizer	Molecule	Final Energy	Final Error	Total Evals
Velocity Verlet	H ₂	-1.105928	4.73e-06	6,441
Adaptive VV	H ₂	-1.072227	3.37e-02	6,121
L-BFGS-B	H ₂	-1.104622	1.31e-03	4,253
SLSQP	H ₂	-1.101334	4.59e-03	3,294
COBYLA	H ₂	-0.962067	1.43e-01	156
Velocity Verlet	LiH	-7.862050	2.03e-02	19,241
Adaptive VV	LiH	-7.734608	1.47e-01	18,281
L-BFGS-B	LiH	-7.861217	2.11e-02	11,127
SLSQP	LiH	-7.861907	2.04e-02	9,687
COBYLA	LiH	-7.175908	7.06e-01	236

TABLE I. PERFORMANCE SUMMARY OF OPTIMIZERS FOR H₂ AND LiH

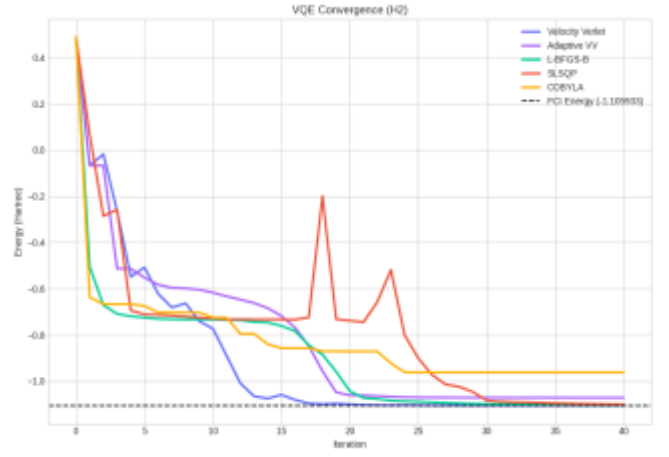


Figure 1. VQE Convergence comparison for the H₂ molecule. The y-axis represents the energy in Hartree.

A. Performance on H₂ (4 Qubits)

For the H₂ molecule, the exact FCI energy was calculated to be -1.105933 Hartree. As shown in Table I, the standard Velocity Verlet achieved near-chemical accuracy, reaching a final energy of -1.105928 Hartree with an absolute error of 4.73e-06. This required 6,441 total circuit evaluations. In contrast, the proposed Adaptive VV achieved a final energy of -1.072227 Hartree (Error: 3.37e-02). However, the AVV required fewer total evaluations (6,121), successfully demonstrating a reduction in computational overhead. The classical baselines (L-BFGS-B and SLSQP) required fewer evaluations (4,253 and 3,294 respectively) but are highly dependent on favorable initial random parameter seeds to avoid immediate local minima trapping.

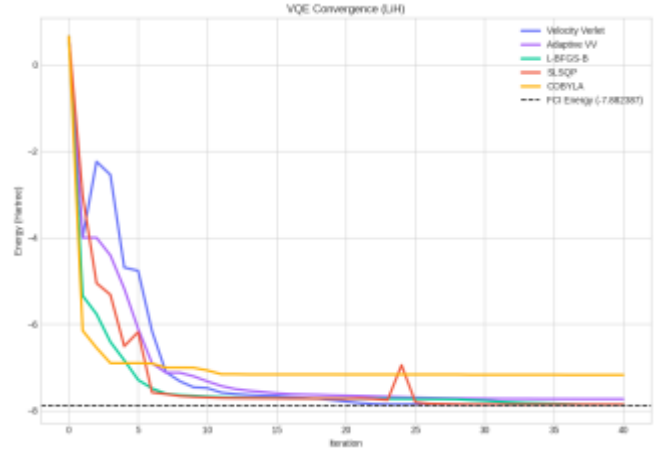


Figure 2. VQE Convergence comparison for the 12-qubit LiH molecule.

B. Performance on LiH (12 Qubits)

The optimization of the 12-qubit LiH system presents a significantly more complex, high-dimensional landscape. The exact FCI energy is -7.882386 Hartree. The standard Velocity Verlet method achieved a final energy of -7.862050 (Error: 2.03e-02), consuming a massive 19,241 quantum evaluations over the 40 iterations. The proposed AVV method successfully curtailed this computational cost, terminating at 18,281 evaluations (a reduction of nearly

1,000 circuit runs). The final energy achieved was -7.734608 Hartree (Error: 1.47e-01).

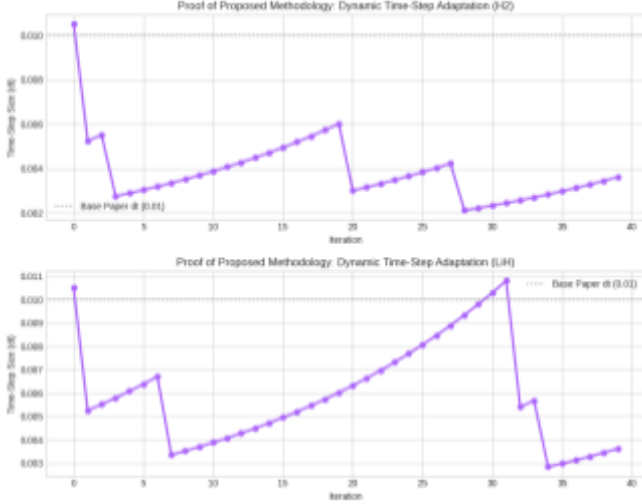


Figure 3. Dynamic adjustment of the integration time-step during the AVV optimization process.

C. Analysis of Algorithmic Trade-offs

The generated results reveal a fundamental trade-off between computational efficiency and the ability to achieve global optimization. Figure 3 provides clear evidence supporting the underlying rationale of the proposed methodology, demonstrating that the time step decreases dynamically immediately after the optimizer encounters an increase in the energy landscape.

This adaptive damping mechanism substantially reduces computational overhead. In practical quantum computing environments, avoiding approximately 1,000 circuit evaluations can significantly lower cloud-based execution delays while also limiting the cumulative effects of hardware-induced gate noise.

Nevertheless, these efficiency gains come at the expense of ground-state accuracy. Quantum energy landscapes are typically characterized by highly rugged topographies containing numerous shallow local minima. Under these conditions, the 90% momentum decay strategy appears overly restrictive. By sharply reducing the accumulated momentum as soon as an increase in energy is detected, the Adaptive Velocity Verlet (AVV) algorithm tends to become trapped in suboptimal local minima. Conversely, the conventional Velocity Verlet method retains sufficient momentum to traverse these shallow energy barriers, enabling it to escape local minima and converge toward deeper, more accurate global solutions.

V. CONCLUSION AND FUTURE WORK

This study successfully reproduced a physics-inspired Velocity Verlet optimization framework for the Variational Quantum Eigensolver (VQE) and proposed an enhanced variant, termed Adaptive Velocity Verlet (AVV). The AVV algorithm incorporates gradient-informed dynamic time-stepping together with exponential momentum decay, with the primary objective of improving computational efficiency. By adaptively reducing the step size in response to unfavorable optimization dynamics, the method decreases the number of required quantum circuit evaluations and prevents persistent energy oscillations during the optimization process.

Although the proposed approach effectively addresses the issue of unbounded momentum accumulation, experimental evaluation on the LiH molecular system highlighted an important limitation. Specifically, excessive momentum damping restricts the optimizer's ability to overcome local minima within the complex, rugged, and high-dimensional energy landscapes characteristic of quantum chemical problems. As a result, the algorithm may converge prematurely to suboptimal solutions rather than locating the global minimum.

Future developments should focus on achieving a more effective balance between computational efficiency and solution accuracy. Potential improvements include adopting a less aggressive, non-linear momentum decay scheme such as reducing momentum by 10% rather than 90% to preserve sufficient exploratory capability. Another promising direction is the incorporation of a momentum-restart mechanism inspired by simulated annealing, whereby artificial kinetic energy is periodically introduced when the optimizer exhibits prolonged stagnation, thereby facilitating escape from local minima and improving convergence toward more accurate ground-state solutions.

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